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Long-range diffusion for weakly camouflaged object segmentation

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ABSTRACT

Weakly supervised camouflaged object segmentation (WSCOS) aims to segment objects well embedded in surroundings via the supervision of sparse annotations. To compensate for the shortcomings of sparse annotations, existing methods design intricate loss functions with multiple regularization rules, not fully exploring the annotation information itself. Therefore, to address this issue, this paper proposes the long-range diffusion network (LRDNet) to diffuse the sparse annotations for improving WSCOS performance. Specifically, a novel gated local saliency coherence (GLSC) loss is designed to efficiently diffuse limited annotation information across the entire image to supplement the supervision by the unidirectional gating. Meanwhile, a two-stage training is introduced to make GLSC loss further improve the diffusion ability of background annotations and then produce the enhanced squeezing effect for sharp edges. Additionally, for capturing sufficient long-range dependencies, the Trans-decorator and the restoration upsampling (RUp) are designed to communicate global priors with convolutional modules by spatial tokens. Extensive experiments have been conducted and the experimental results demonstrate the effectiveness and the high versatility of our designed LRDNet. The code is available at https://github.com/Ray3417/LRDNet.

1. Introduction

The goal of camouflaged object segmentation (COS) is to identify objects that are visually embedded in the background, meeting the requirements of practical applications such as military object segmentation (Gou et al., 2023; Liu & Di, 2023), medical image segmentation (Bo et al., 2023; Fan et al., 2020b), and agriculture pest detection (Liu et al., 2019; Rizzo et al., 2023). COS is more challenging compared to other object segmentation tasks due to the high similarity between the foreground and the background in terms of color, texture, etc. To address this challenge, different strategies, such as bio-inspired strategy (Fan et al., 2020a; Mei et al., 2021; Wang et al., 2024; Yan et al., 2021), multi-source information fusion strategy (Lin et al., 2023; Wang et al., 2023b; Xiang et al., 2021; Zhong et al., 2022), and multi-task strategy (Le et al., 2019a; Lv et al., 2021a; Zhao et al., 2023), are employed to assist in segmenting the camouflaged objects.

Though existing COS methods have achieved superior performance, they generally rely on fully supervised learning and require large-scale datasets with pixel-level annotations, which is time-consuming and costly. To alleviate this issue, He et al. introduced the weakly supervised learning (WSL) into COS task and proposed a weakly supervised camouflaged object segmentation (WSCOS) method (He et al., 2023). They trained their model with subsets of annotated pixels such as points or scribbles, leveraging the accuracy of these annotations to improve the performance of WSCOS. However, the sparsity of point and scribble annotations limits the model's learning capacity (He et al., 2023).

As we all know, to compensate for the shortcomings of sparse annotations, loss function design by utilizing multiple regularization rules is an effective way of improving the performance of WSL. Loss design in WSL aims to utilize multiple regularization rules to compensate for the shortcomings of sparse annotations. Some methods (Li et al., 2023; Obukhov et al., 2019; Yu et al., 2021) construct loss functions by introducing the cross-pixel consistency into regularization to supplement the supervision information. These loss functions utilize the similarity between image pixels to roughly constrain the structural inference of predicted mappings based on the assumption (Pixels with similar features or closer distances are more likely to belong to the same class). In addition, some methods (Xu et al., 2022; Yu et al., 2021; Zhang et al., 2020) design the loss functions by introducing the cross-view consistency into the regularization, and then they maximize the similarity between

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predictions of the original task and the pretext tasks (random rotation, random scaling, etc.) based on the assumption (A good segmentation model should have consistent predictions for the same objects before and after transformations).

Through analysis and research, we find that some existing loss functions based on the cross-pixel consistency (such as gated conditional random fields (GCRF) loss Obukhov et al., 2019, local saliency coherence loss (LSC) Yu et al., 2021, and context affinity loss (CA) He et al., 2023) do not fully utilize the annotation information even if they work with the partial cross-entropy loss. Specifically, the annotated pixel positions of predicted mapping first receive supervision information from the ground truth (point, scribble) and approach the annotation values (pCE). Then, the loss functions minimize the difference between predicted values of similar pixels to diffuses the supervision information globally. However, the interaction between similar pixels is mutual, that means the annotated pixels influence the similar unannotated pixels, meanwhile the unannotated pixels can also affect the similar annotated pixels. So the surrounding unlabeled similar pixels preventing the predicted values of annotated pixels from quickly approaching their ground truth. Furthermore, this interference disrupts the global propagation of annotation information. Therefore, to avoid bidirectional influence we design a novel loss with the unidirectional gating for efficiently diffusing the sparse annotations in this paper. Therefore, we design a novel loss to efficiently diffuse the sparse annotations by the unidirectional gating in this paper.

Apart from the design of the loss function, the design of the model itself also has a certain impact on the performance of WSL. Although convolution is widely used in computer vision (Wei et al., 2015; Zhang et al., 2023, 2022), pure convolutional architectures still have some limitations, such as short distance modeling. In WSCOS, the existing methods (He et al., 2024, 2023) are generally pure convolutional architectures, which will lead to a limited receptive field and reduce the diffusion accuracy. Long-range modeling capability facilitates accurate estimation of diffusion scales and reduces diffusion cycles. To address this limitation, we propose a novel network architecture that integrates token-replacement self-attention mechanisms with classic convolutional modules. This framework synergistically captures global dependencies through token-wise interactions while maintaining local feature extraction capabilities. The proposed design further incorporates dynamic threshold control and feature alignment mechanisms to effectively guide annotation propagation toward camouflaged regions.

In summary, we conduct an in-depth study on the weakly supervised camouflaged object segmentation (WSCOS) with sparse annotations (points and scribbles), and design a novel long-range diffusion network (LRDNet), which focuses on loss design and model design to improve the segmentation performance of the model. In terms of loss design, a novel loss function, namely gated local saliency coherence loss (GLSC), is designed to diffuse the sparse annotations by unidirectional gating with high efficiency. Additionally, a second stage training is introduced to make our designed GLSC loss improve the diffusion ability of background annotations and then produce the enhanced squeezing effect[1] for sharp object edges. The diagram of the diffusion and squeeze process are shown in Fig. 1. Notably, the unidirectional approximation makes the second backpropagation mainly come from object edges, which help avoid overfitting and error accumulation issues that often arise in multi-stage training (Tang et al., 2018a,b). In terms of model design, inspired by Mobile-Former (Chen et al., 2022), we design a Trans-Decorator module to update and store global priors with minimal tokens, meanwhile, the exchange of local and global information is facilitated by the lightweight cross-attention. Additionally, to address the loss of spatial information caused by downsampling, we design the restoration upsampling (RUp) module for feature alignment across different resolutions. We conducted extensive experiments and the results demonstrate that our proposed LRDNet can significantly improve the segmentation performance compared to the state-of-the-art WSCOS methods. In addition, LRDNet is also applied into the weakly supervised salient object segmentation (WSSOS) task and it outperforms existing WSSOS methods in most metrics.

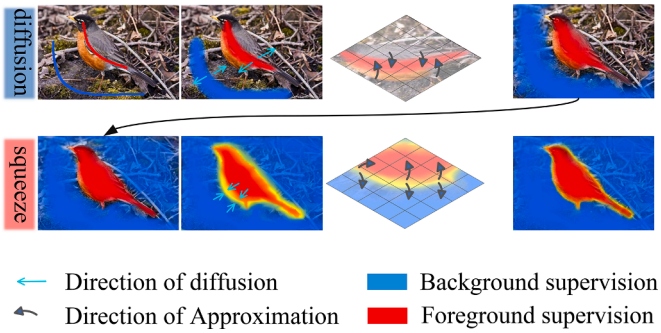


Fig. 1. Schematic diagram of the diffusion & squeeze process. In the first row, sparse annotations are efficiently diffused according to pixel similarity; In the second row, the generated prediction is continued to spread, producing a squeezing effect at the edge of camouflaged object.

Our contributions can be summarized as follows:

- We propose a novel long-range diffusion network (LRDNet) for the WSCOS task. It can capture sufficient long-range dependency and enhance the supervised information diffusion.
- A local saliency coherence loss (GLSC) is designed to facilitate the diffusion of the sparse annotations by restricting the unidirectional approach between annotated and non-annotated pixels. With the second training stage, GLSC can also enhance the squeezing effect to force the clear edges.
- A Trans-Decorator module is designed to update and store global priors with minimal tokens and exchange local and global information via the lightweight cross-attention. Additionally, a restoration upsampling module (RUp) is designed to align features of different resolutions.
- Extensive experimental results indicate that GLSC outperforms the state-of-the-art WSCOS methods, meanwhile our method also achieves superior results on the WSSOS task.

2. Related work

2.1. Camouflaged object segmentation

Camouflaged object segmentation (COS) focuses on identifying the undetectable objects and has made significant progress in recent years. According to the label category, the existing COS methods can be roughly divided into fully supervised COS (FSCOS) and WSCOS.

In term of FSCOS, by imitating the process of animal hunting and human observation, some methods (Fan et al., 2020a; Jia et al., 2022) based on biological inspiration are proposed to improve the model's understanding of camouflaged objects. Some methods supplement the limited RGB information by introducing depth (Wang et al., 2023a; Wu et al., 2023) and frequency domain (Lin et al., 2023; Zhong et al., 2022) to adapt to challenging camouflage environments. Considering CNN-based methods are difficult to fully capture global contextual information, some methods (Hu et al., 2023; Liu et al., 2023a; Xing et al., 2023) introduce transformer as backbone in COS. Huang et al. (2023) observed the limited effectiveness of local feature modeling in the transformer backbone and proposed the FSPNet, which progressively shrinks and decodes locally enhanced neighboring transformer features. These transformer-based COS methods have demonstrated that global

[1] The foreground and background annotations diffuse throughout the entire image via the loss function, causing the boundary between them to narrow, which is referred to as the squeezing effect.

information contributes to great localization and understanding of camouflage environments. However, the introduction of transformer is computationally expensive, leading to inefficient training and inference processes. These mentioned methods, whether based on CNN or transformer, rely on fully supervised learning and require large amount of training data with pixel-level annotations, which is time-consuming and costly.

To address this issue, weakly supervised camouflaged object segmentation (WSCOS) has emerged and has become a research hotspot in recent years. CRNet (He et al., 2023) is the first WSCOS method to train COS models with annotated scribble datasets, but sparse annotations inevitably led to poor segmentation performance. WS-SAM (He et al., 2024) proposed the generating pseudo-labels to improve the discriminative ability of the model. Although WS-SAM achieved good segmentation performance, it relies on the large model SAM (Kirillov et al., 2023) for accurate and stable guidance, reducing its practicality.

Therefore, to address the aforementioned issues of existing WSCOS methods, we propose a novel WSCOS method to boost the camouflaged object segmentation performance.

2.2. Weakly supervised segmentation

Weakly supervised segmentation aims to achieve the segmentation of the object through low-cost labels. Existing studies are mainly conducted from label design and loss design.

In terms of label design, some studies use bounding boxes (Liu et al., 2021; Oh et al., 2021) or image-level labels (Piao et al., 2021; Zeng et al., 2019) to supervise models for pixel-level predicted maps. The main strategy of these methods is to design class activation maps (CAM) that map image-level labels to pseudo-label masks, which can guide the model's focus on the region of interest. These two types of labels can provide rich semantic information, but their supervision signals are highly uncertain. Annotations at the subset level of annotated pixels, including points (Gao et al., 2022a,b) and scribbles (Lee et al., 2021; Li et al., 2024; Wang et al., 2023c), have been widely used in weakly supervised learning. The annotated parts with these labels have high reliability but their sparsity weakens the regularization ability of the model.

Thus, some losses were designed for the issues of sparse annotations. Normalized cut loss and kernel cut loss (Tang et al., 2018a) were proposed for regularizing scribble semantic segmentation. Gated structure-aware loss (Zhang et al., 2020) was designed to enhance consistency within prominent regions and alleviate the blurring issue caused by smooth losses (Godard et al., 2017; Wang et al., 2018). Based on GCRF (Obukhov et al., 2019), Yu et al. (2021) proposed the local saliency coherence loss (LSC), which gradually diffuses annotation information throughout the whole image with cross-pixel consistency. However, these losses fail to make full use of the sparse annotation information. And the approximation between annotated pixels and non-annotated pixels is bidirectional, which greatly reduces the efficiency of annotations propagation.

Therefore, we design a new loss with a unidirectional approximation and a two-stage training for the sparse annotations to improve the WSCOS performance.

3. Proposed methodology

This section introduces our proposed weakly supervised camouflaged object segmentation method, namely Long-Range Diffusion Network (LRDNet). Firstly, we give a brief overview of our method; Secondly, we present the details of Trans-Decorator (TD) and Restoration Upsampling (RUp) in LRDNet; Lastly, we introduce our designed Gated Local Saliency Coherence (GLSC) loss in details, as well as the feasibility and effectiveness of its relevant two-stage characteristics.

3.1. Overview

The overall architecture of our method is shown in Fig. 2, which includes the network pipeline LRDNet, loss group, and the training strategy. For the WSCOS task, the total dataset consists of the sparsely annotated training dataset $D_{train} = \{T_i, S_i\}_{i=1}^{N}$ and testing dataset $D_{test} = \{X_i, Y_i\}_{i=1}^{N}$, where T_i represents the ith training image and X_i represents the ith test image, Y_i denotes the ith pixel-level ground truth and N is the size of the dataset. $S_i \in \{0, 1, 2\}$ denotes the sparse annotations, where 1, 2, and 0 represent the foreground, the background, and the unknown pixels respectively. For the actual training, S_i is mapped to $\{0, 1, 255\}$, where 1, 0, and 255 represent the foreground, the background, and the unknown pixels respectively.

3.2. LRDNet architecture

Firstly, given an input image $I \in \mathbb{R}^{3\times H\times W}$, ResNet50 is adopted to extract features and then their channels are unified to generate the backbone features $F_k, k \in \{1, 2, 3, 4\}$, where k represents the kth level backbone features. Secondly, we adopt two classic convolutional modules,

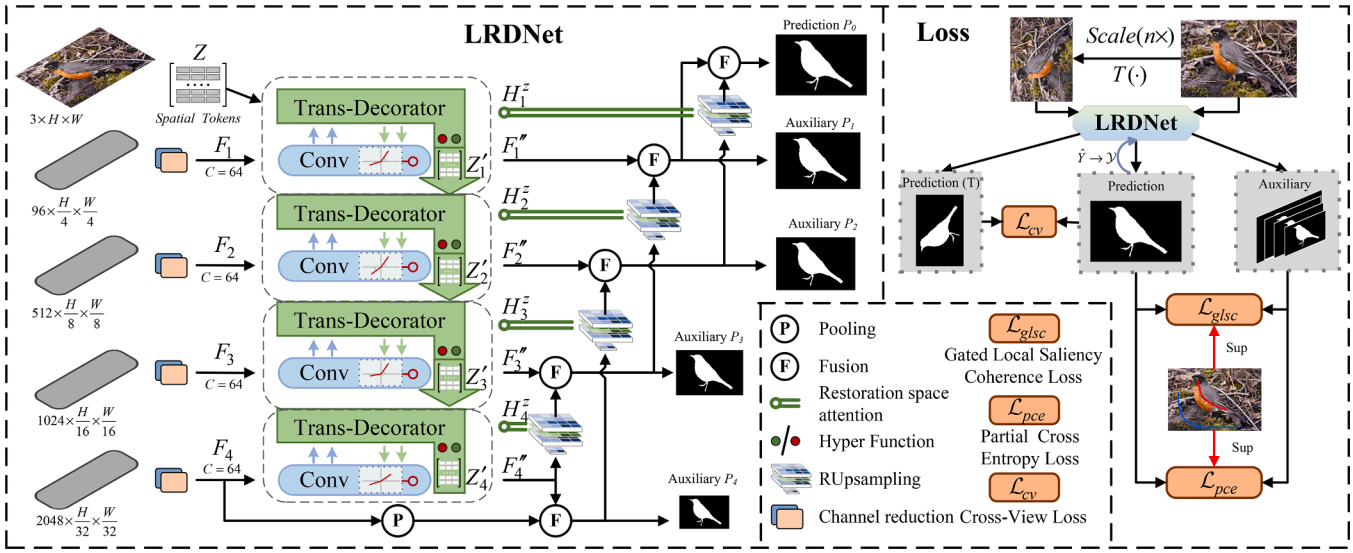


Fig. 2. An overview of our proposed method. The left figure is the architecture of LRDNet while the right figure illustrates the loss group.

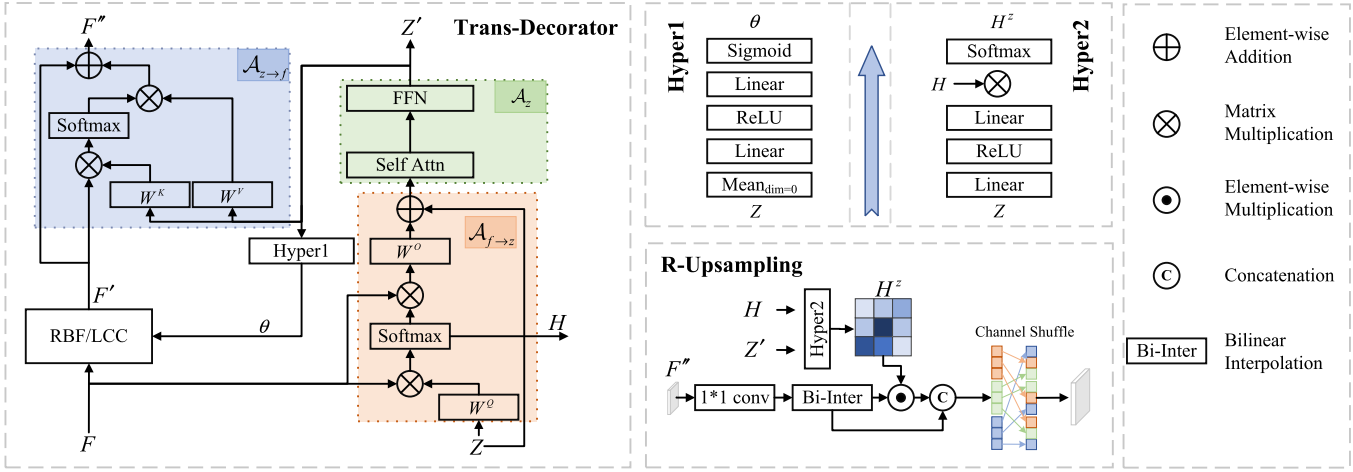


Fig. 3. The architecture of Trans-Decorator and RUpsampling.

receptive field block (RFB) (Wu et al., 2019) and local-context contrasted block (LCC) (He et al., 2023) (see appendix for detailed structure diagram), both enhanced with Dy-ReLU (Chen et al., 2020), to extract the semantic features F_3' and F_4' from the high-level features F_3 and F_4 respectively and the structural features F_1' and F_2' from the low-level features F_1 and F_2 respectively. Thirdly, our designed Trans-Decorator can be attached to these above convolutional modules to enhance global perception. Trans-Decorator takes backbone features F_k, local features $F_k', k \in \{1,2,3,4\}$ (semantic features and structural features), and learnable spatial tokens $Z \in \mathbb{R}^{M\times d}$ as the inputs, where M and d is the number and dimension of the tokens respectively. These spatial tokens Z can learn and record the global priors from the features at the same level. Then tokens Z are fused with F_k and F_k' in the form of Dy-ReLU parameters and space attention to generate globally enhanced features F_k'', restoration space attention H^z, and updated tokens Z'. In addition, we design the restoration upsampling (RUp) to align the space feature across different resolutions. Finally, the pooled F_4 is fused with the globally enhanced features F_k'' according to the cross-aggregation strategy (Zhao et al., 2021) to generate the predicted mask sequences P (including main output P_0 and the auxiliary outputs P_1 to P_4).

3.3. Loss group

In order to efficiently address the sparse annotation problems, the gated local saliency coherence (GLSC) loss $\mathcal{L}_{glsc}$ with two-stage characteristics is designed. Based on existing research (He et al., 2023; Yu et al., 2021), the classical partial cross-entropy loss (pCE) $\mathcal{L}_{pce}$ and the consistency loss $\mathcal{L}_{cv}$ with a auxiliary task $\mathcal{T}$ are adopted to cooperate with our GLSC ($\mathcal{L}_{glsc}$) to consist of the loss group of our model. In addition, to fully exploit the two-stage characteristics of the GLSC, we design a two-stage training approach, where the output of the first-stage training is used as the soft pseudo-labels for supervising the second-stage training. During both rounds of training, the main output P_0 participates in the calculation of all three losses while the auxiliary outputs $P_1 - P_4$ only participate in the calculation of the GLSC loss and the pCE loss.

Next, we will focus on introducing the designed Trans-Decorator with restoration upsampling (RUp) and GLSC loss in detail.

3.4. Trans-decorator & restoration upsampling

Existing WSCOS models only capture the local feature dependencies with small perception, making it difficult to learn global information from the limited annotation information. To address this issue, this paper proposes the LDD network consist of a Trans-decorator and a restoration upsampling (RUp). The Trans-decorator captures global information from the backbone features and generates spatial tokens which are the key judgment basis for the parameters θ of the Dy-ReLU. Furthermore, the spatial tokens are mapped to low-complexity spatial weights to participate in lightweight cross-attention calculations. Finally, with the global information retained by these tokens, and restoration upsampling can generate restoration space attention to effectively supplement the information loss caused by downsampling.

3.4.1. Trans-decorator

The detailed structure of the Trans-Decorator is shown in Fig. 3. This module has three inputs: (a) backbone features F_k, (b) local features F_k', and (c) spatial tokens $Z \in \mathbb{R}^{M\times d}$, randomly initialized by the embedding vectors.

Firstly, a simplified cross-attention $\mathcal{A}_{f\to z}(F, Z; W^Q, W^O)$ is designed to fuse the backbone features F_k and the spatial tokens Z. While the weights $\mathcal{W} = softmax(\frac{QK^T}{\sqrt{d_k}})$ of this cross-attention are outputted to participate in the restoration upsampling.

Then, a standard transformer module $\mathcal{A}_z(Z; W^Q, W^K, W^V)$ is used to spatially weight the tokens and obtain updated tokens Z' with global priors. Specifically, the simplified cross-attention $\mathcal{A}_{f\to z}(F, Z; W^Q, W^O)$ and the standard transformer module $\mathcal{A}_{f\to z}(F, Z; W^Q, W^O)$ can be described as follows:

$$\mathcal{A}_{f\to z}(F, Z; W^Q, W^O) = \left[Attn(z_i W_i^Q, f_i, f_i)\right]_{i=1:h} W^O \tag{1}$$

$$\mathcal{A}_z(Z; W^Q, W^K, W^V) = FFN\left\{\left[Attn(z_i W_i^Q, z_i W_i^K, z_i W_i^V)\right]_{i=1:h}\right\} \tag{2}$$

where the backbone features F_k and spatial tokens Z are split into h heads as $F_k = [f_1 \cdots f_h]$, $Z = [z_1 \cdots z_h]$ for multi-head attention. The split for the ith head $z_i \in \mathbb{R}^{M\times \frac{d}{h}}$ is different to the ith token $z_i \in \mathbb{R}^d$. W_i^Q, W_i^K, and W_i^V are the query, key, and value projection matrix for the ith head respectively. W^O is used to combine multiple heads together and $Attn(Q, K, V)$ is the standard attention function. $[\cdot]_{1:h}$ denotes the concatenation of h elements and $FFN\{\cdot\}$ is the feed-forward network. Secondly, the updated tokens Z' are used to compute the parameters $\theta = f(z'_m)$ to control the threshold of Dy-ReLU for the nonlinear mapping of local features F_k'. Here z'_m is the average of all global tokens and $f(\cdot)$ consists of two layers of MLP with ReLU activation (as shown in Fig. 3). Further, Z' is fed into a simplified cross-attention $\mathcal{A}_{z\to f}(F', Z'; W^K, W^V)$ to generate globally-aware features F'':

$$\mathcal{A}_{z\to f}(F', Z'; W^K, W^V) = \left[Attn(f'_i, z'_m W_i^K, z'_m W_i^V)\right]_{i=1:h} \tag{3}$$

where the local features F_k' are split into h heads as $F_k' = [f_1' \cdots f_h']$. The Trans-Decorator outputs the globally enhanced features $F_k'', k \in \{1,2,3,4\}$, the updated tokens Z' and cross-attention weights H, where Z' is used as the inputs for the next level.

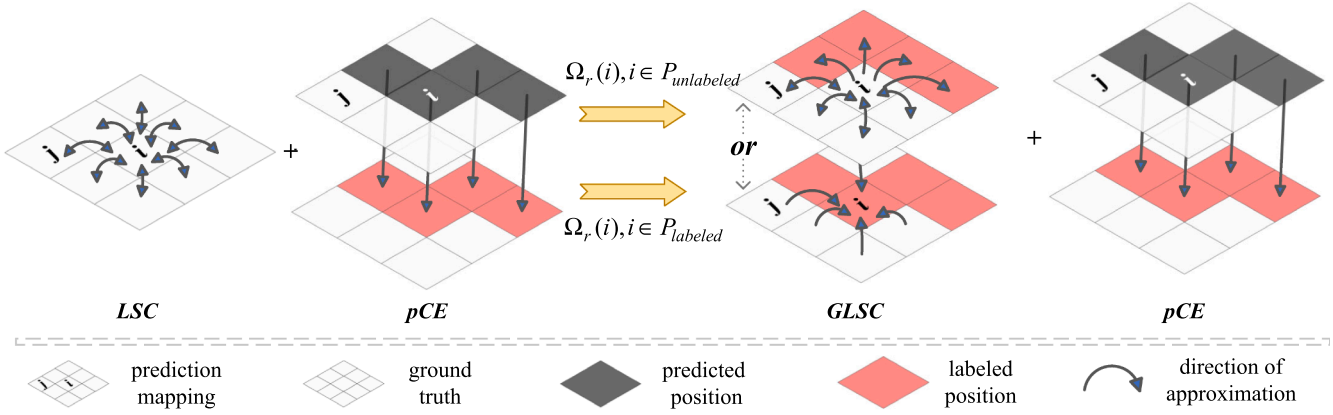


Fig. 4. Principle comparison of LSC and GLSC. The pink area is the labeled location, and the gray area is the predicted location.

3.4.2. Restoration upsampling

The obtained globally-aware features F_k'' for each level are fused via the cross-aggregation strategy. However, directly adopting parameter-free upsampling leads to the feature misalignment problem (Huang et al., 2021). Hence, we design the restoration upsampling (RUp) illustrated in Fig. 3 to align features across different resolutions. The RUp has three inputs: attention weights H, update tokens Z' and the globally-aware features F_k'' to be sampled.

We utilize attention weights H and update tokens Z' to generate restoration space attention $H_z = softmax(g(f(Z'), H))$, where $g(\cdot)$ represents the matrix multiplication and $f(\cdot)$ consists of two layers of MLP with ReLU activation. Then the globally-aware features F_k'' are upsampled and dot multiplied with H_z to achieve spatial feature alignment. In addition, to promote inter-channel information exchange, we randomly shuffle the aligned feature channels. Therefore, the process of restoration upsampling (RUp) can be written as:

$$RUp(F_k'', H_z) = [C_{1\times1}(Bi(F_k'' \cdot H_z)) + F_k'']_{g=c} \tag{4}$$

where $C_{1\times1}$ is the 1×1 convolution layer with batch normalization and ReLU activation function, $Bi(\cdot)$ is the bilinear interpolation, and $[\cdot]_{g=c}$ denotes the random channel shuffle with c groups.

3.5. Loss function

In order to achieve high quality weakly supervised predictions, we propose an effective loss function group, including the gated local saliency coherence (GLSC) loss, the partial cross-entropy (pCE) loss, and the cross-view loss (Yu et al., 2021). Specifically, for the WSCOS, we analyze and improve the classical LSC loss from approximation direction and squeeze effect aspects.

3.5.1. LSC & direction of approximation

The LSC is based on the manifold assumption that pixels with similar features tend to have the similar class probabilities ($L_1 \to 0$). The pixel similarity, evaluated by the gated conditional random field (Obukhov et al., 2019), weights L_1 norm to force consistent prediction of class for similar pixels. The specific calculation process is as follows:

$$\Psi(I, \hat{y}) = \sum_{a=1}^{N} \sum_{\substack{b=1 \\ b\neq a}}^{\Omega_r(a)} K_{ab} L_{1_{ab}} \tag{5}$$

where $\Omega(a) = [a_x - r, a_x + r] \times [a_y - r, a_y + r]$ is a neighbor window with radius r of a position $a \in [1, N]$ with coordinates (a_x, a_y). b represents one position within the window except a. It should be noted that K_{ab} is calculated on image I, while L_1 is calculated on the prediction $\hat{y}$. K_{ab} represents the weighted sum of pixel feature similarity for P kernels, and it can be written as:

$$K_{ab} = \sum_{p=1}^{P} w^{(p)} k^{(p)}(f_a^{(p)}, f_b^{(p)}) \tag{6}$$

$$k^{(p)}(f_a^{(p)}, f_b^{(p)}) = \frac{1}{\omega} \exp\left(-\frac{\left\| f_a^{(p)} - f_b^{(p)} \right\|^2}{2\sigma_p^2}\right) \tag{7}$$

where $f_a^{(p)}, p \in \{color, position\}$ denotes the color feature (RGB) or the position feature (coordinate) vector, $w^{(p)}$ is the weight of different pixel features, σ is the hyperparameter, $\|\cdot\|^2$ is the L_2 operation, and $1/\omega$ is the normalized weight.

With the partial cross-entropy (pCE) loss, LSC can diffuse annotation information to the entire image, as shown in Fig. 4. However, minimizing the norm approximation is bidirectional, that means $a \to b$ and $b \to a$. This limits the annotation information propagation. Therefore, we design a mask m_{src} to fully utilize annotation information for achieving the efficient partial unidirectional approximation. The rules for the mask m_{src} are as follows:

$$L_1|m_{src} = \begin{cases} |s_a - \hat{y}_b| & a \in labled, b \in unlabled \quad i \\ |s_b - \hat{y}_a| & a \in unlabled, b \in labled \quad ii \\ |\hat{y}_a - \hat{y}_b| & a \in unlabled, b \in unlabled \quad iii \\ |s_a - s_b| & a \in labled, b \in labled \quad iv \end{cases} \tag{8}$$

where $s \in \{0, 1\}$ represents the value of effective annotations and $|\cdot|$ represents the absolute value.

In fact, m_{src} can decouple LSC and pCE. In the case of i and ii, the original collaboration between LSC and pCE is chaotic because $\hat{y}_i, i \in labled$ has two pathways for backpropagation, one from $\hat{y}_j, j \in \Omega_r(a)$ (noise) and another from s_i. By removing the calculation related to $\hat{y}_i, i \in labled$ from LSC, the parameter update process is optimized.

By adding a gate mechanism to Eq. (5), we achieve the unidirectional improvement of LSC, the gated local saliency coherence (GLSC) loss, which can be written as:

$$\mathcal{L}_{glsc}(I, \hat{y}) = \sum_{a=1}^{N} \sum_{\substack{b=1 \\ b\neq a}}^{\Omega_r(a)} K_{ab} L_{1_{ab}} | m_{src} \tag{9}$$

3.5.2. GLSC & squeeze effect

One of the flaws in sparse annotations is the inability to provide accurate edge supervision signals for addressing the challenges of blurred edges in COS. Therefore, we propose a configuration of GLSC in two-stage training to improve the quality of edge prediction by leveraging the enhanced squeezing effects (as shown in Fig. 1).

For the second training stage, the predictions $\hat{Y}$ generated in the first training stage are utilized as the pseudo-labels $\mathcal{Y}$, and the other settings

remain the same as those in the first training stage. The mapping rules between $\mathcal{Y}$ and $\hat{Y}$ are as follows:

$$\mathcal{Y}_i = \begin{cases} \hat{Y}_i & if \quad \hat{Y}_i = 1 || \hat{Y}_i = 0 \\ 255 & otherwise \end{cases} \tag{10}$$

The squeeze effect is not significance on LSC because it relies on passive diffusion[2] to diffuse supervision information, which has limited assistance in the propagation of background annotations. The two-stage training with the GLSC loss enable the active diffusion, which centers around the effective annotations $\mathcal{Y}_i \in \{0, 1\}$ and spreads to the surrounding areas. Active diffusion enhances the squeeze effect in two ways: (a) the squeeze effect between background annotations and foreground annotations will appear around the foreground because the active diffusion is not restricted by the similar pixels; (b) the squeeze effect is amplified due to the larger area of annotations and smaller spacing between the effective annotations. It is worth noting that GLSC only applies the squeeze effect to the edge regions as shown in Eq. (8), which effectively prevents the overfitting problem and the error accumulation problem. With the influence of the passive diffusion and the active diffusion, uncertain edges are compressed and narrowed, leading to the improvement of the edge prediction quality.

3.5.3. Loss group

According to previous studies (He et al., 2023; Yu et al., 2021), self-consistency mechanism can enhance the multi-scale detection capability of networks. Therefore, we introduce auxiliary tasks (rotation, scale) and cross-view loss $\mathcal{L}_{cv}$ (He et al., 2023) to further constrain the model predictions. The cross-view loss $\mathcal{L}_{cv}$ can be expressed as:

$$\begin{aligned} \mathcal{L}_{cv}(\hat{y}, \tilde{y}) &= (1+\lambda)\mathcal{L}_{cv'}(\hat{y}_d, \tilde{y}) + (1-\lambda)\mathcal{L}_{cv'}(\hat{y}, \tilde{y}_d) \\ \mathcal{L}_{cv'}(\hat{y}, \tilde{y}) &= \frac{1}{M}\sum_i (1-\alpha) \cdot \frac{1-SSIM(\hat{y}_i, \tilde{y}_i)}{2} + \alpha \left| \hat{y}_i - \tilde{y}_i \right| \end{aligned} \tag{11}$$

where $SSIM$ is the single scale SSIM (Godard et al., 2017), $\tilde{y}$ is the prediction of the auxiliary task. M is the total number of pixels and i is a pixel index. α is set to 0.85 and λ is set to 0.3. $\hat{y}_d$ and $\tilde{y}_d$ represents non-gradient $\hat{y}$ and $\tilde{y}$ respectively.

The total objective function of our proposed model LRDNet can be written as:

$$\mathcal{L} = \mathcal{L}_{cv} + \beta \mathcal{L}_{glsc} + \mathcal{L}_{pce} \tag{12}$$

where β is set to 0.3 in this paper.

4. Experiments

4.1. Experiment setup

4.1.1. Datasets

In the experiments, we use three COS datasets, including CAMO (Le et al., 2019b), S-COD (He et al., 2023), and NC4K (Lv et al., 2021b). 3040 images from S-COD and 1000 images from CAMO are used as the training set, as the same time the remaining images are used as the testing dataset.

4.1.2. Metrics

Following existing methods (Fan et al., 2022, 2020a), four common metrics, including mean absolute error (MAE) (Perazzi et al., 2012), adaptive F-measure (F_β) (Margolin et al., 2014), mean E-measure (E_m) (Fan et al., 2018), and structure measure (S_α) (Fan et al., 2017) are adopted in our experiments for evaluation. Smaller MAE, or larger F_β, E_m, and S_α means better performance.

[2] Passive diffusion refers to the propagation method relying on similar pixels in the image. The background is usually complex with multiple groups of similar pixels. The background annotations only cover a small number of groups of similar pixels, thus passive diffusion is hard to diffuse background annotations.

4.1.3. Implementation details

Our proposed LRDNet is implemented by PyTorch and trained on Geforce GTX 1080Ti. The ResNet-50 pre-trained on ImageNet-1K (Deng et al., 2009) is used as the backbone to extract the original features. The optimizer is AdamW with Beta parameters set to 0.5 and 0.999. In addition, we adopt the fixed step-size decay (StepLR) strategy to adjust the learning rate. Here the learning rate is set to 1e-4, the decay period is set to 100, and the decay rate is 0.5. The main scale of the images is set to 384 pixels.

4.2. Comparison with state-of-the-arts

In order to verify the effectiveness of our proposed LRDNet, we conduct extensive experiments on CAMO, NC4K, and S-COD datasets respectively and give the quantitative and qualitative evaluation in this section.

4.2.1. Quantitative evaluation

LRDNet is compared with 5 WSCOS methods, including SAM (Kirillov et al., 2023), WSSA (Zhang et al., 2020), SCWS (Yu et al., 2021), TEL (Liang et al., 2022), and CRNet (He et al., 2023), meanwhile it is compared to two fully supervised COS models, SINet (Fan et al., 2020a) and BASNet (Qin et al., 2021). The results of quantitative evaluation are shown in Table 1. For WCOS task, point and scribble are utilized as the sparse annotations in these experiments.

From Table 1 we can see that: 1) For WSCOS task with the point as the sparse annotations, our proposed method LRDNet outperforms the second-best method CRNet in all metrics on the CAMO dataset. On the other two datasets, LRDNet also achieves an average improvement of 5.18 % in E_m and 8.57 % in M. 2) For WSCOS task with the scribble as the sparse annotations, our proposed method LRDNet achieves SOTA in all metrics on three datasets. In particular, LRDNet achieves an average improvement of 5.18 % in F_β and 2.9 % in S_α on the S-COD dataset. 3) Compared to fully supervised COS methods, our proposed LRDNet can achieve similar performance and even outperform them on certain metrics. For example, on the CAMO dataset, LRDNet achieves an average 4.2 % improvement in E_m over SINet. 4) In addition, LRDNet is compared with CRNet*, the CRNet with our loss group instead of its original complex loss group. CRNet* also achieves improvement in almost all metrics on scribble datasets. In particular, CRNet* outperforms CRNet by 2 % in E_m on NC4K. These results indicate that our loss group with greater simplicity can help our model achieve better performance than the original loss group.

4.2.2. Qualitative evaluation

To intuitively show the effect of camouflage object segmentation, visual comparison of our LRDNet, SINet, SCWSSOD, and CRNet is conducted and the results are shown in Fig. 5.

It can be seen that: 1) In the first row, our LRDNet can capture the part of the tail better compared with CRNet. 2) In the second row, for dark background, our LRDNet can generate accurate edges with higher confidence than SINet. 3) In the third row, compared to SINet and CRNet, our method can identify the fracture position well. 4) In the last row, LRDNet can effectively alleviate the edge smoothing problem, and even can generate sharp edges similar to the fully supervised COS models.

In general, LRDNet can identify the complete structure of camouflaged objects and accurately generate high-quality object edges. It can be mainly attributed as follows: 1) The designed Trans-Decorator is able to generate the convenient global information supplementation; 2) with the two-stage training, our designed GLSC loss can enhance the squeezing effect to force the clear edges.

Table 1
Quantitative results on three benchmarks with point supervision and scribble supervision. CRNet* indicates the original CRNet with our loss group. The best two weakly-supervised results are in **bold** and *italic* fonts.

Models	*CAMO*				*NC4K*				*S-COD*			
	$M\downarrow$	$S_\alpha\uparrow$	$E_m\uparrow$	$F_\beta\uparrow$	$M\downarrow$	$S_\alpha\uparrow$	$E_m\uparrow$	$F_\beta\uparrow$	$M\downarrow$	$S_\alpha\uparrow$	$E_m\uparrow$	$F_\beta\uparrow$
Full Supervision												
BASNet(Arxiv2022)	0.096	0.749	0.796	0.696	0.058	0.817	0.859	0.767	0.038	0.802	0.855	0.707
SINet(CVPR2020)	0.092	0.745	0.804	0.712	0.058	0.808	0.871	0.768	0.043	0.776	0.864	0.667
Point Supervision												
SAM(Arxiv2023)	0.16	0.643	0.639	0.597	0.118	0.717	0.723	0.675	0.093	0.73	0.737	**0.673**
WSSA(CVPR2020)	0.148	0.649	0.652	0.607	0.104	0.743	0.756	0.688	0.087	0.642	0.733	0.509
SCWS(AAAI21)	0.142	0.687	0.672	0.624	0.098	0.754	0.767	0.695	0.082	**0.738**	0.777	0.593
TEL(CVPR22)	0.133	0.645	0.674	0.662	0.085	*0.766*	0.795	0.725	0.063	*0.727*	0.803	*0.623*
CRNet(AAAI23)	0.137	0.663	0.688	0.629	0.08	0.758	0.796	**0.744**	*0.06*	0.711	0.802	0.607
CRNet*	*0.112*	*0.72*	*0.812*	*0.666*	**0.071**	**0.768**	*0.855*	0.707	0.062	0.717	*0.81*	0.585
our	**0.103**	**0.722**	**0.818**	**0.679**	**0.071**	0.765	**0.857**	*0.729*	**0.057**	0.72	**0.829**	0.614
Scribble Supervision												
SAM(Arxiv2023)	0.160	0.643	0.639	0.597	0.118	0.717	0.723	0.675	0.093	0.730	0.737	0.673
WSSA(CVPR2020)	0.118	0.696	0.786	0.615	0.091	0.761	0.779	0.657	0.071	0.684	0.770	0.536
SCWS(AAAI21)	0.102	0.713	0.795	0.658	0.073	0.784	0.814	0.723	0.055	0.710	0.805	0.602
TEL(CVPR22)	0.104	0.717	0.797	0.681	0.075	0.782	0.832	0.754	0.057	0.724	0.826	0.633
CRNet(AAAI23)	0.092	0.735	0.815	0.709	0.064	0.779	0.853	*0.751*	0.049	0.733	0.832	0.637
CRNet*	**0.088**	*0.741*	*0.822*	*0.711*	*0.062*	*0.787*	*0.870*	0.750	*0.048*	*0.745*	*0.837*	*0.644*
our	*0.091*	**0.747**	**0.838**	**0.719**	**0.060**	**0.792**	**0.873**	**0.757**	**0.047**	**0.754**	**0.843**	**0.669**

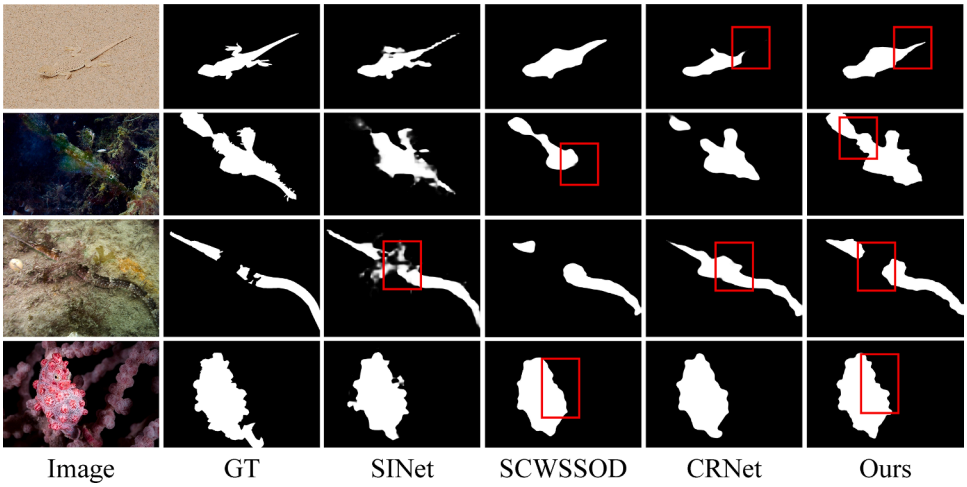


Fig. 5. Visual comparison between our method and other state-of-the-art methods.

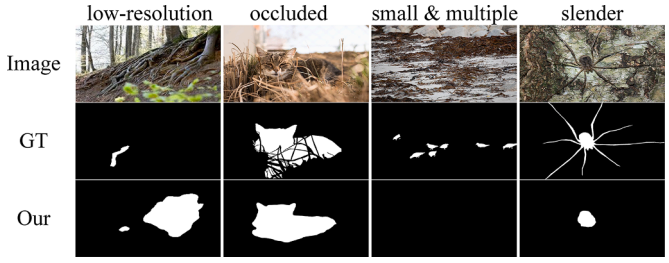


Fig. 6. Illustration of failure cases.

4.3. Failure cases

Although our LRDNet can accurately segment camouflaged objects in most cases, it is still not robust enough for some extremely challenging cases, as shown in Fig. 6. Specifically, in the first column, our LRDNet cannot accurately locate the camouflaged object with low-resolution. In the second column and the last column, our method encounters challenges in identifying extremely narrow occlusions or narrow objects. However, other models tend to produce the same or worse prediction results in these cases.

These failures can be attributed to: too cluttered backgrounds leading to confusion, low-resolution images with limited information, and lack of training samples.

4.4. Ablation studies

In this paper, the Trans-decorator, the RUp and the GLSC-based loss group are designed for WSCOS. To verify their effectiveness, we conduct the ablation studies of these three modules with the supervision of scribble annotations on S-COD dataset.

4.4.1. Ablation studies on modules

To verify the effectiveness of each module, four ablation experiments are conducted on the S-COD dataset. Keeping the loss strategy unchanged, we adopt the classic convolutional component of LRDNet as the baseline. Because of the dependence of RUp on Trans-Decorator, in

Table 2
The ablation study results of components on S-COD.

No.	Baseline	TD	RUp	$M \downarrow$	$S_\alpha \uparrow$	$E_m \uparrow$	$F_\beta \uparrow$
i	✓			0.048	0.745	0.837	0.644
ii	✓	✓		**0.047**	0.751	0.841	0.662
iii	✓		✓	0.048	0.749	0.839	0.651
iv	✓	✓	✓	**0.047**	**0.754**	**0.843**	**0.669**

this ablation experiment removing the Trans-Decorator module refers to cut off information exchange between the Trans-Decorator and convolutional modules. The detailed experimental results are shown in Table 2. It can be seen that:

Firstly, the Trans-Decorator modules improve the performance of the model in all metrics (ii). For example, S_α and F_β increase by 0.9 % and 2.7 %. This fully proves that the iterative spatial tokens can provide effective long-range information to improve the segmentation ability of the model. Secondly, the addition of RUp module improves baseline by 1.1 % according to F_β (i, iii). It can be attributed to the ability of the RUp module to make use of spatial tokens to repair the loss of spatial information of the corresponding resolution features. Thirdly, the model achieves the best performance when two modules are added simultaneously (iv). It demonstrates the synergy of our Trans-Decorator and RUp module.

4.4.2. Ablation studies on loss functions

In this section, we conduct ablation studies on loss functions with different stages. The ablation studies on combination of three loss functions L_{cv}, L_{glsc}, and L_{pce} are executed respectively in the 1 stage. In addition, the comparison of L_{glsc} and L_{lsc} are given in the first and second stages respectively, as well as the combination of L_{cv}, L_{glsc}, and L_{pce} is conducted in the third stage. The experimental results are shown in Table 3.

The results indicate that all three loss functions significantly contribute to the performance. Specifically, the combination of L_{glsc} and L_{pce} (iii) outperforms both individual L_{pce} (i) and the combination of L_{pce} and L_{cv} (ii). This demonstrates the synergistic effect between pCE and GLSC designed in our work. The results of ablation between LSC and GLSC demonstrate that our designed GLSC outperforms LSC in both the first and second stages (iv, v, vi, vii). Especially, GLSC shows significant improvement over the origin LSC in the second stage, indicating its powerful edge refinement capability. Furthermore, the third stage training (viii) is tried and the results indicate that no further performance growth over the second stage training. This may be because that the second stage primarily focuses on optimizing predictions with lower confidence region such as edges.

4.5. Further analysis

In this section, we further analyze our model LRDNet and the segmentation results, including edge analysis, generalization analysis, and hyperparameter setting, to demonstrate the generality of LRDNet and the rationality of its interpretation.

Table 3
The ablation study results of loss functions on S-COD.

No.	Stage	Setting	$M \downarrow$	$S_\alpha \uparrow$	$E_m \uparrow$	$F_\beta \uparrow$
i	1	w/ pce	0.155	0.622	0.641	0.425
ii		w/ pce, cv	0.073	0.679	0.699	0.546
iii		w/ pce, glsc	0.061	0.677	0.743	0.576
iv		w/ pce, cv, lsc	0.050	0.741	0.834	0.654
v		w/ pce, cv, glsc	0.048	0.749	0.832	0.662
vi	2	w/ pce, cv, lsc	0.049	0.745	0.833	0.655
vii		w/ pce, cv, glsc	**0.047**	**0.754**	**0.843**	**0.669**
viii	3	w/ pce, cv, glsc	0.046	0.752	0.834	0.660

4.5.1. Edge analysis

The designed GLSC loss and its second-stage expansion aim to optimize edge detection from two aspects: confidence and fine structure. Here we give the visual comparison of edge structures and the visual comparison of confidence regions at the edges in Fig. 7 respectively.

In Fig. 7(a), our model LRDNet with 1 and 2 stages is compared to CRNet respectively. It can be observed that our method LRDNet outperforms other two methods by significantly reducing the area of low-confidence regions, achieving the clear and accurate edge predictions. In Fig. 7(b), our method LRDNet is compared to CRNet on fine structure. We can see that the predicted results of our model exhibit significant non-smooth features at complex edges, which are the results of repeated squeezing during the diffusion process of foreground-background annotations.

4.5.2. Generalization analysis in WSSOS

We have demonstrated the effectiveness of our proposed LRDNet in the WSCOS task. However, it is also expected to work effectively in WSSOS task. To verify this, on four SOS datasets ECSSD (Yan et al., 2013), HKU-IS (Li & Yu, 2015), THUR (Cheng et al., 2014), and DUTS-TEST (Wang et al., 2017) we conduct experiments with scribble annotations, as well as compare our LRDNet to five WSSOS methods, including WSSA (Zhang et al., 2020), SCWSSOD (Yu et al., 2021), NSAL (Piao et al., 2022), CAGNet (Liu et al., 2023b), and CGRNet (Liang et al., 2024).

The experimental results, as shown in Table 4, indicate that our method can achieve performance comparable to those state-of-the-art WSSOS methods. Specifically, on the ECSSD dataset, our LRDNet improves by 14.3 % and 2.7 % in M and E_m respectively; on the DUTS-TEST dataset, LRDNet improves by 18.4 % and 3.3 % in M and E_m respectively. These experimental results demonstrate that our proposed LRDNet has good generalization.

4.5.3. Hyperparameters analysis

In order to analyze the effect of some hyperparameters (pseudo label mapping threshold, token length, token dim, and loss weight β) on our LRDNet model, we conducted experiments on the S-COD dataset with different hyperparameter configuration schemes and the results are shown in Table 5.

For the mapping threshold of pseudo-labels, when it is set to (1,0), the model can achieve the best performance in all metrics. This indicates that stricter screening conditions mean better results, because this is able to alleviate the error accumulation problem to some extent. For the token length and the token dimension, the model works best when they are set to 3 and 128 respectively. Note that the token dimension 192 results in an additional computational cost with limited performance. For the loss weight β, it is set to 0.1, 0.3, and 0.6 respectively with the classic configuration (He et al., 2023) in this experiment. The experimental results indicate that when the loss weight β is set to 0.3, the model can achieve better performance compared with the model with β set to 0.1, meanwhile, it can obtain the relatively performance compared to the model with β set to 0.6. Therefore, for the fairness in comparison, loss weight β is set to 0.3 in our model and the related experiments.

4.5.4. Analysis of computational cost

In this section, the analysis of computational cost have been given with comparison with the baseline CRNet. Here MACs, Params, and Speed are adopted as the metrics and the compared results are shown in Table 6.

From Table 6, it can be seen that: the computational cost of our model LRDNet is equivalent to that of the compared CRNet according to MACs, Params, and Speed. Contrasting with CRNet's reliance on 64-channel, our LRDNet adopts a compact 32-channel pathway to realize feature communication for compensating the computational cost, caused by the long-range dependency modeling analogous to attention mechanisms. Therefore, the proposed LRDNet can obtain substantial

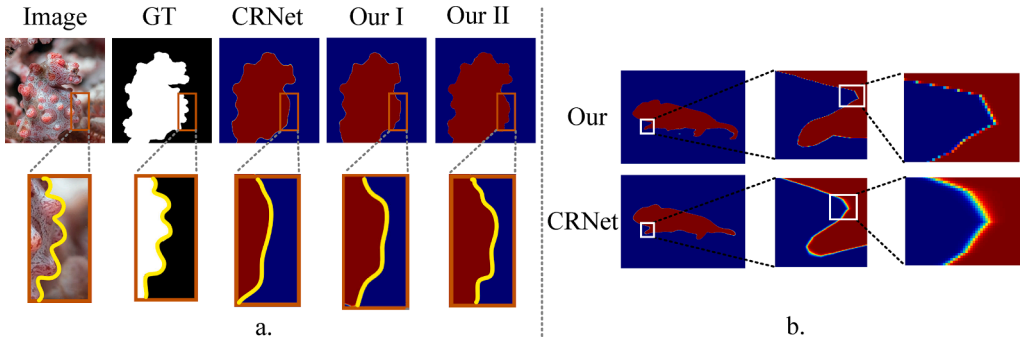


Fig. 7. Visual comparison of edge about structures (a.) and confidence regions (b.).

Table 4
Quantitative results on saliency benchmarks with weakly supervised learning. The best two results are highlighted in **BOLD** and *ITALIC*. 'F_β^α' means adaptive F-measure.

model (year)	ECSSD				HKU-IS				THUR				DUTS-TEST			
	$S_\alpha\uparrow$	$M\downarrow$	$E_m\uparrow$	$F_\beta^\alpha\uparrow$	$S_\alpha\uparrow$	$M\downarrow$	$E_m\uparrow$	$F_\beta^\alpha\uparrow$	$S_\alpha\uparrow$	$M\downarrow$	$E_m\uparrow$	$F_\beta^\alpha\uparrow$	$S_\alpha\uparrow$	$M\downarrow$	$E_m\uparrow$	$F_\beta^\alpha\uparrow$
WSSA (2020)	0.863	0.061	0.908	0.865	0.865	0.047	0.923	0.858	0.8	0.077	0.837	0.718	0.803	0.062	0.865	0.747
SCWSSOD (2021)	0.879	0.051	0.908	0.9	0.882	0.038	0.938	0.896	*0.815*	*0.069*	*0.845*	**0.754**	0.841	0.049	*0.89*	0.822
NSAL (2022)	0.825	0.082	0.882	0.853	0.849	0.053	0.92	0.859	_	_	_	_	0.778	0.074	0.849	0.729
CAGNet (2023)	0.864	0.056	0.908	0.878	0.851	0.05	0.923	0.854	_	_	_	_	0.793	0.06	0.865	0.75
CGRNet (2024)	*0.89*	*0.049*	*0.914*	*0.88*	**0.911**	*0.033*	*0.942*	*0.892*	_	_	_	_	**0.864**	*0.049*	0.876	**0.834**
our	**0.897**	**0.042**	**0.939**	**0.908**	*0.901*	**0.031**	**0.951**	**0.899**	**0.825**	**0.068**	**0.855**	**0.754**	*0.86*	**0.04**	**0.919**	**0.834**

Table 5
Hyper-parameter analysis on S-COD.

	Pseudo label mapping threshold			Token length			Token dim			Loss weight β		
Metrics	(0.8, 0.2)	(0.9, 0.1)	(1, 0)	1	3	6	64	128	192	0.1	0.3	0.6
$M\downarrow$	0.051	0.048	**0.047**	0.048	**0.047**	**0.047**	0.049	**0.047**	**0.047**	0.049	**0.047**	0.048
$S_\alpha\uparrow$	0.750	0.752	**0.754**	0.752	**0.754**	**0.754**	0.748	0.754	**0.755**	0.747	**0.754**	0.753
$E_m\uparrow$	0.833	**0.843**	**0.843**	0.844	0.843	**0.845**	0.828	**0.843**	0.842	0.839	**0.843**	0.842
$F_\beta\uparrow$	0.658	0.663	**0.669**	0.664	**0.669**	0.667	0.651	**0.669**	0.668	0.667	0.669	**0.670**

Table 6
Comparison of computational costs between CRNet and our LRDNet.

Models	MACs (G)	Params (M)	Speed (fps)
CRNet	17.02	32.28	31.52
our	16.56	32.13	30.08

performance improvements (as quantified in Section 4.5.1) without introducing additional computational cost.

5. Conclusions

In this paper, we are committed to alleviate the sparsity problem of the WSCOS annotations. From the perspective of diffusion annotation, we propose a novel long-range diffusion method LRDNet, which includes a GLSC loss, the Trans-decorator, and a RUp module. Firstly, the GLSC loss is designed to efficiently diffuse annotation information by the unidirectional gate. Meanwhile, the two-stage training is further set for GLSC to enhance the diffusion of background annotations and make the object edges sharp. Moreover, to accurately diffuse the annotation information we design the Trans-decorator, which can capture the long-range dependence and communicate global priors by few spatial tokens. Lastly, the designed RUp can introduce these spatial priors to align features at different levels. We evaluate our method on the benchmark datasets with the supervision of point annotations and scribble annotations and compare it with various state-of-the-art methods qualitatively and quantitatively. Extensive experimental results illustrate the superiority of our method and further indicate its effectiveness and generalization.

CRediT authorship contribution statement

Rui Wang: Writing – original draft, Formal analysis, Conceptualization, Investigation; **Caijuan Shi:** Conceptualization, Writing – review & editing; **Weixiang Gao:** Software; **Changyu Duan:** Supervision; **Ao Cai:** Visualization; **Fei Yu:** Writing – review & editing; **Yunchao Wei:** Supervision.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Visualization of intermediate model results

(Fig. A.1).

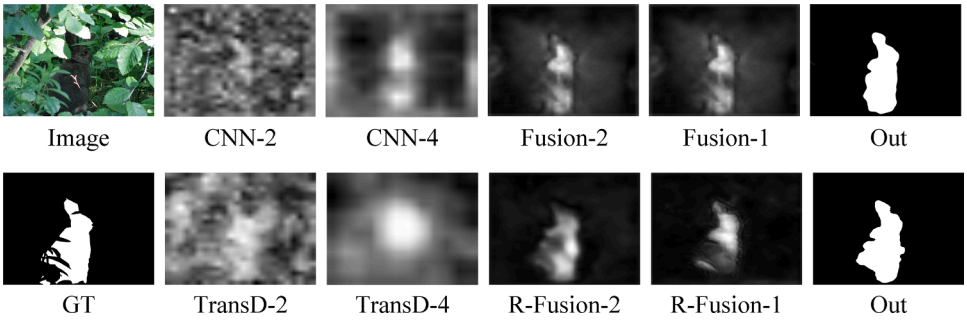


Fig. A.1. Visualization of intermediate model results. Specifically, "CNN" is the output of CNN module, and "Trans-D" is the output of our Trans-Decorator module of the n-th stage. "R-Fusion" and "Fusion" is the output of our fusion module and its no RUp version respectively. "-n" denote the nth level.

Appendix B. Structures of RFB-modify and LFE

(Fig. B.1).

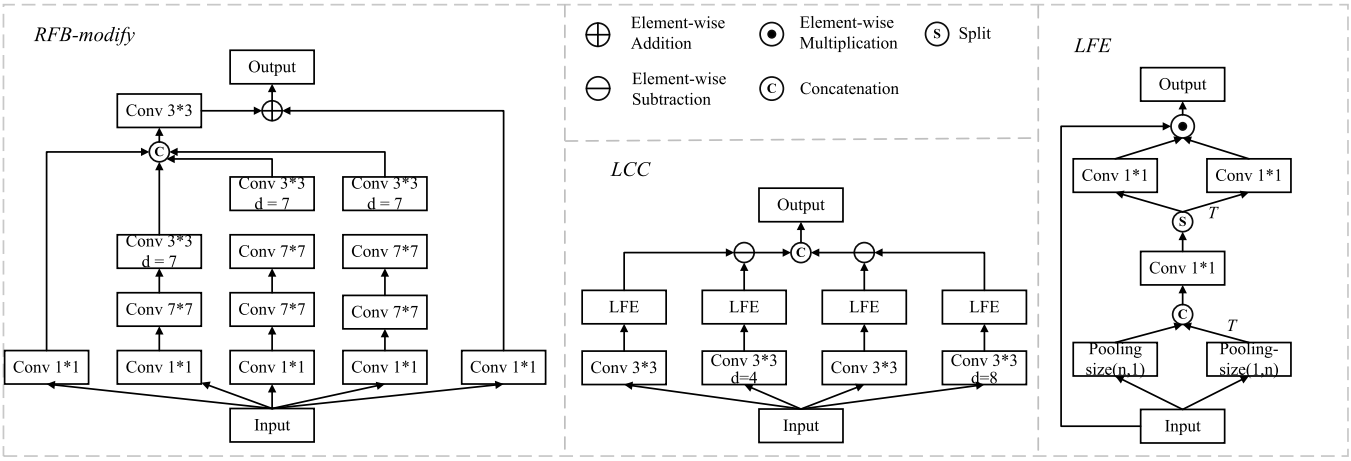


Fig. B.1. Structures of RFB-modify and LFE.

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